

Assignment 4: FO for FPA

Part 1: Feed at the center of the focal plane of the parabolic reflector

Write a MATLAB code to analyze in reception an antenna located in the center of the focal plane of a parabolic reflector. The geometry is shown in Fig. 1(a). Suppose that the antenna can be modeled as a circular feed with a linearly x-polarized uniform amplitude distribution. The operating frequency is 220 GHz. The aperture of the circular feed has a diameter of $D_f = 3.6\lambda_0$. The parabola has a diameter of $D_r = 130\lambda_0$ and f-number $f_{\#} = 1.8$.

Assume that the parabolic reflector is illuminated by an x-polarized plane wave with an incident angle of $\theta_i = 0^\circ$ and $\phi_i = 0^\circ$ with amplitude of $E_0^{PW} = 1V/m$.

1.1 Calculate the GO electric field on the equivalent FO sphere of this geometry.

1.2 Evaluate the electric far field of the circular feed over the same FO sphere. Plot this field together with the GO field in the $\phi = 0^\circ$ and 90° planes of the FO sphere.

1.3 Using the analysis of antennas in reception for a generalized incidence, calculate the power received at the input terminals of the feed, assuming impedance matched condition. Provide the value of the aperture efficiency of the overall reflector antenna system.

Assume now that incident plane waves are illuminating the reflector with the following incident angle range: $\theta_i = [0: 3^\circ]$ and $\phi_i = [0: 360^\circ]$.

1.4 Calculate the GO electric fields on the equivalent FO sphere of this geometry for each incident plane wave angle.

1.5 Plot the power received by the reflector feed, in dBs and normalized, as a function of this incident angle. This plot represents the radiation pattern of the reflector antenna in reception mode. Compare this plot with the far field radiation pattern calculated using a transmission analysis of the same structure, as developed in MATLAB assignment 2.

1.6 Compare the aperture efficiencies, directivities and gains obtained using both Tx and Rx analysis.

Part 2: Feed displaced in the focal plane of the parabolic reflector

Now consider that the same circular feed is displaced in the focal plane by a quantity d_x as indicated in Fig. 1(b).

2.1 Consider $d_x = D_f = 3.6\lambda_0$. Plot the pattern in reception of this displaced field and evaluate the aperture efficiency. Compare them to the feed in the center. What is the pointing direction in the displaced case? What differences do you notice between the two beams? Argue the cause for these differences (if any).

2.2 How does the aperture efficiency and pointing direction of the displaced feed changes in the following cases:

a) $d_x = D_f = 2\lambda_0 f_{\#}$ (the case discussed in 2.1),

b) $d_x = D_f = \lambda_0 f_{\#}$, and

c) $d_x = D_f = 0.5\lambda_0 f_{\#}$?

What are the beam crossing levels in dB for each of the three sampling scenarios above? Plot the central and the neighboring beam, compare these beams between the three cases and their crossing level against the theoretical expected values. In case of discrepancies between obtained and theoretical values of the beam crossing, explain the cause for this difference.

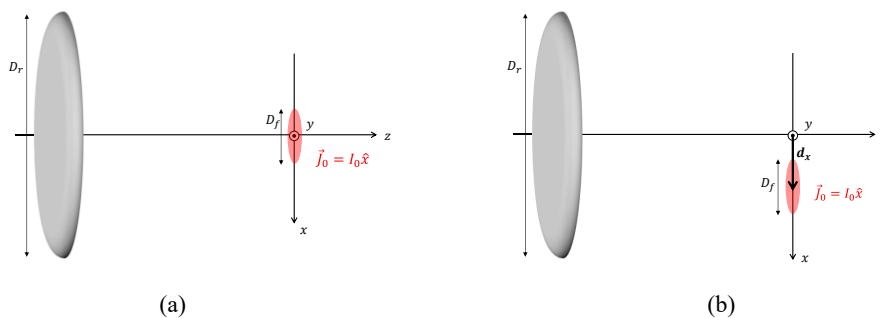


Fig. 1: The scenario with feed (a) at the center of the focal plane, and (b) displaced.